SSU Data Dash 2026: *Team name*

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Video (YouTube, unlisted is fine): `https://youtu.be/YOUR-VIDEO`
Code and submission files (GitHub): `https://github.com/YOUR-TEAM/YOUR-REPO`

Summary

We answer two questions with two models, and we do not mix the scores. A desk that can see the listing gets a close: walk-forward median absolute percentage error 4.28%, against 7.26% for taking the ask as the forecast, with the typical prediction still 0.9% high. A desk marking a home before anyone has published an ask gets a value: walk-forward MdAPE 7.45%, against 15.33% for a 12-month community dollars-per-square-foot anchor. Off-market counts double. The list price correlates 0.986 with the sold price, so the tighter number is the ask.

The accuracy is in the columns and in one calibration, not in a retuned tree. Both models are one LightGBM on a log target, with the same settings, fit only on sales that have already closed. The list-aware model predicts the gap around the ask. A regression and a SARIMAX turn the market's drift into a column, and the tree decides the pass-through. The off-market model builds the level from a local tape, same-building sales brought forward to the listing month, and a remarks score fit out of time, then applies one shift from the prior three months. The failures point opposite ways. The list-aware model is rich on cheap homes. The off-market model is cheap on expensive homes, and cheap again when the tape goes stale.

1 Data and features

The book is 150,000 synthetic Toronto sales, January 2021 through May 2026. Names and coordinates do not match real maps, so we join nothing. Outside data: none. Both models see the same 10,069 validation sales, December 2025 through May 2026, and both are refused any sale that had not closed yet. An inner slice, July through November 2025, is where we froze the windows, the phrases, the building rule, and the shift weight. Those months then go back into the estimation sample. The walk-forward is the number we quote, because it never scores a month the tree has seen.

Every model sees log floor area (the midpoint when the listing wrote a range), a lot-missing flag, log frontage, depth, beds, baths, parking, garage, latitude, longitude, the listing month as a sine and cosine, and type, style, municipality, and community. Lot size is missing on every condo and present on every house, so the flag is a house-versus-apartment cut. A cell with fewer than 30 past sales falls back, community to type to city. An unseen category stays missing. The equations for the columns below are in Appendix A and Appendix B.

1.1 List-aware

Column	What it is
Log ask, lot area, area-was-a-range	The ask, and a flag when size was written as a range.
Peer ask	List $/sqft divided by median sold $/sqft, same community and type, closed before the listing month.
Three-month discount	Median sold/list in that cell over the three prior sale months. Missing on 46% of a frozen fit.
City lag	Median log gap in the previous calendar month.
Month factor	SARIMAX forecast of the monthly mean residual from a log-linear fit on the ask.

Split gain on the fitted tree: peer ask 22%, community 19%, log ask 10%, city lag 9%, three-month discount 8%, month factor 6%. The lot-missing flag is 0.07%, because the ask has already named the product. Remarks, tried as a text decomposition, moved developmental MdAPE from 4.83% to 4.85%, so this track has no text (Appendix D.1).

1.2 Off-market

Column	What it is
12-month community $/sqft	Median sold $/sqft, same community and type, prior 12 months. The level the ask would have carried.
City $/sqft, 3 and 12 months	The same median, citywide, at two windows.
Building price and count	Same pin and type, closed before listing, reindexed to the listing month. Present for 60% of validation homes.
Thirteen phrase flags	Written patterns in the remarks. Not a fitted model.
Remarks score	A ridge of log price on the description. Each training row is scored by earlier sales only.

Split gain: lot missing 33%, building price 18%, log size 13%, community 12%, log frontage 8%. The remarks score is 0.7%. The comps are doing the work. Municipality is about zero once community is in the tree.

2 Models

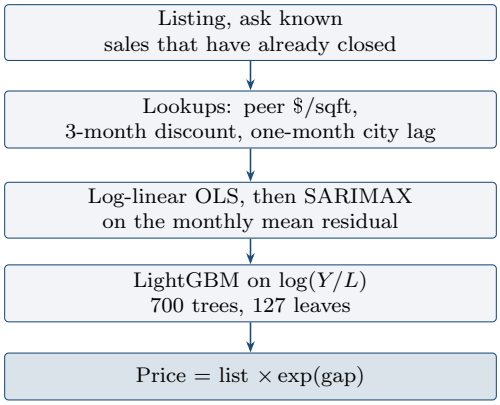
Same tree, two targets. LightGBM, squared error, 127 leaves, learning rate 0.03, 700 trees, at least 20 rows in a leaf, row and column subsamples 0.8, L2 penalty 1, seed 0. We did not reopen that grid after the list-aware target became the log gap. Retuning an earlier specification on July–November and then refitting produced a December–May MdAPE of 4.57%, worse than the untuned 4.53%.

2.1 List-aware

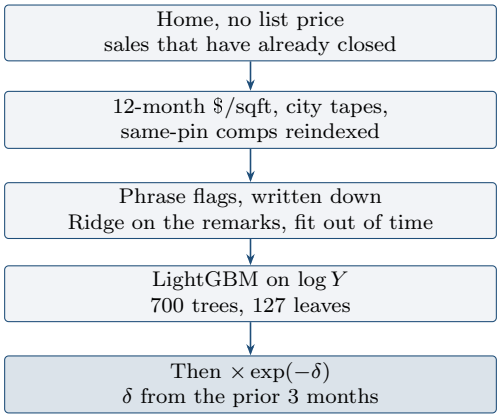
The target is log(sold/list). The price is the ask times exp(gap). The month factor is a column, 6.3% of gain, and the tree is the last step. A level shift after the tree won one frozen window and then sat 1.75% low on the walk. We did not adopt it (Appendix D.1).

2.2 Off-market

The target is log(sold). The price is then multiplied by $\exp(-\delta)$ at full weight. δ is the median log miss on the three months before the origin, from a tree that did not see those months, and the walk re-estimates it every month. On the July–November screen, no shift was MdAPE 8.31 and +4.71% high. Full weight was 7.97 and +3.77%. On the one-origin fit, $\delta = +1.58\%$. A model that starts from the community anchor reached walk-forward MdAPE 7.50. The direct model is ahead in five months, and February is the loss (Appendix C). We kept the direct model because it won the screen and then held the walk.



List-aware. The SARIMAX is one column. The tree is the last step.



Off-market. Flags are patterns. The ridge is a model. δ is the only second stage.

Figure 1: Scoring path for each subsection above. The list-aware chain ends at the tree. The off-market chain has one shift after it.

3 Validation

The walk is one procedure for both tracks. For each month from December 2025 through May 2026, every table, the SARIMAX, the ridge, the categories, and the tree are rebuilt on sales that closed before that month. The off-market refit also rebuilds δ on the three months before the scored month. The month is then scored. Its sold prices are in none of those objects, and on the off-market track the list price is hidden. A building comp still has to predate the home's listing. The July–November choices stay fixed. Pooled MdAPE weights homes equally, so May (2,291 sales) outweighs January (1,058). A random split leaks later closes into the peer medians and the building comps. `run.py validate` is this procedure. The month path is Appendix C.

4 Results

Track	Homes	MdAPE % (95% CI)	MAPE % (95% CI)	Median signed error %
off-market	10,069	7.45 (7.29–7.63)	9.79 (9.61–9.97)	-0.37
list-aware	10,069	4.28 (4.20–4.39)	5.37 (5.28–5.46)	+0.90

Table 1: Walk-forward, December 2025–May 2026, 10,069 homes. Brackets: bootstrap 95% interval, a lower bound because one month moves together.

Handout bars we did not refit: 17.7%, 20.8% for ten nearest sales, and 7.3% for the list price (our refit 7.26%). The hidden test runs through September 2026 and should land near this table. We did not choose either model on these six months.

4.1 List-aware

The model beats the ask in every month. MdAPE stays inside 3.97 to 4.55. Community is worth 0.08 points over the same model without it (4.36%), and it is ahead in every month. The month-by-month comparison with the ask is Figure 3 in the appendix.

4.2 Off-market

The model cuts the 12-month community anchor from 15.33% to 7.45%. The pooled signed error is −0.37%. That pool is the average of six months that do not share a sign. The comparison with the anchor, month by month, is Figure 4 in the appendix.

5 Where the model fails

The cheap end is already a bit high on both tracks, so one citywide percent cannot fix either tail.

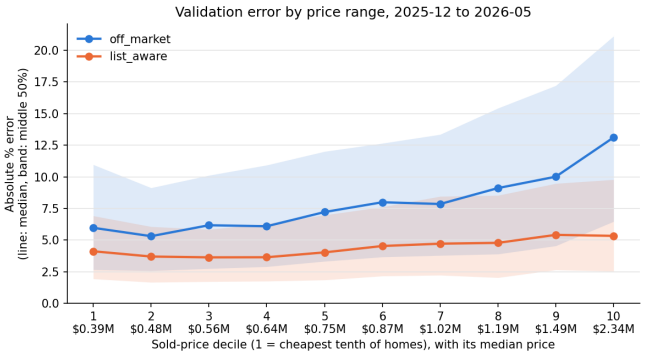


Figure 2: Walk-forward absolute percentage error by sold-price decile. The line is the median. The band is the middle half of homes.

5.1 List-aware

On a one-origin fit the model is 3.7% high on homes under about $440,000, and close to unbiased in the top decile (MdAPE 5.6%). Signed error stays positive every month, from +0.20% to +1.33%. Detached and semi-detached cost about two extra points of MdAPE. Across municipalities the error stays inside 4.2% to 4.6%.

5.2 Off-market

The top decile, from about $1.8 million, has MdAPE 13.6% and is about 11% low. Signed error changes sign and reaches −4.44% in February (Appendix C). Detached and semi-detached cost about three to four extra points of MdAPE, and municipalities stay inside 6.3% to 7.7%. Scoring May 2026 with the tape cut off at the end of January moves MdAPE from 7.39% to 8.38% and the signed error from +0.98% to −5.29%.

6 Limitations

Both outputs are a point price. The book is synthetic, the claim stops in May 2026, and the test files run to September.

6.1 List-aware

On the one-origin window the 95th percentile of absolute error is about 14%, against 19% for the ask. We would not add a haircut. The walk-forward already rejected one (Appendix D.1).

6.2 Off-market

The 95th percentile is about 26%, against 57% for the community anchor. A mark above $1.5 million, or on a tape older than about 90 days, should carry that limit next to the number.

Reproducing our results

From a fresh copy of the repository:

```
pip install -r requirements.txt
python run.py validate
python run.py report
python run.py predict
python run.py check
```

`validate` writes the walk-forward file. `report` writes the table and the figure above. `predict` refits on every closed sale through May 2026, so the off-market shift is then measured on March–May 2026, and writes the two submission files. `check` confirms one positive finite price per test id.

Appendix

The numbered report ends at Limitations. This appendix is separate from those four pages. It holds the formulas (Appendix A, Appendix B), the month path and the baseline charts (Appendix C), and the specifications we measured and did not keep (Appendix D.1, Appendix D.2).

Appendix A List-aware formulas

Peer ask, for a home listed in month τ, with floor area S and ask L:

$$\text{peer ask} = \frac{L/S}{m_{c,t}(\tau)},$$

where $m_{c,t}(\tau)$ is the median sold dollars per square foot in the same community and type among sales that closed before τ. Fewer than 30 sales uses the type. The community cell covers 97% of the developmental validation rows. This is 22% of list-aware gain.

The month factor starts from a log-linear fit of log sold price on log ask and the attributes. On the estimation sample the ask elasticity is 0.894 and R^2 is 0.966. We take the monthly mean residual. An augmented Dickey–Fuller test with a trend gives $p = 0.0002$, so we do not difference. The fitted model on the 59 estimation months is a constant, a one-month autoregression, and a 12-month autoregression: intercept -0.0046, AR(1) 0.872, seasonal AR(12) 0.370. In-sample R^2 on the monthly series is 0.885. Putting that forecast back onto the log-linear prediction, as a hard adjustment, cuts developmental MdAPE from 7.13% to 4.78% and the median signed error from +5.95% to +0.25%. The tree receives the same number as a column and uses it at a 6.3% gain share. The median value on the developmental validation rows is -0.0539.

Appendix B Off-market formulas

The community tape, window $W = 12$, is the median sold dollars per square foot in the same community and type over the prior W months, then type, then city. We froze 12 months on July–November: MdAPE 15.33, against 16.19 for 6 months, 15.70 for 24, and 16.42 with signed error +11.97% for the full history. The community cell itself is used for 76% of developmental validation rows.

A building comp shares the pin, latitude and longitude rounded to six decimals plus property type, closed strictly before the listing date, and is a different listing. Each historical dollars-per-square-foot is stored relative to a type index that uses only earlier months, a three-month median of monthly median log dollars per square foot. At the listing month the comps are brought back to that month's index, and the building price is size times their median. The gate is one prior sale. Requiring three was worse on July–November.

The remarks ridge is log sold price on a bag of unigrams and bigrams, penalty 0.3, at most 3,000 terms, minimum document frequency 40. Numbers and community names are stripped before the vectorizer runs, so the score cannot memorise a printed price or a neighbourhood. Inside the estimation sample each row is scored from sales that closed earlier. July–November preferred the score by 0.05 MdAPE, 7.97 against 8.02.

Appendix C Walk-forward by month

The two tracks are scored on the same homes, so the months stay in one table. A positive signed error means the prediction is high.

		List-aware		Off-market	
Month	Homes	MdAPE	Signed	MdAPE	Signed
2025-12	1,238	3.97	+1.31	7.42	+0.63
2026-01	1,058	4.16	+1.33	7.27	−2.50
2026-02	1,442	4.38	+0.20	8.17	−4.44
2026-03	1,816	4.14	+0.46	7.10	−1.15
2026-04	2,224	4.37	+1.31	7.44	+1.65
2026-05	2,291	4.55	+0.92	7.39	+0.98
Pooled	10,069	4.28	+0.90	7.45	−0.37

C.1 List-aware

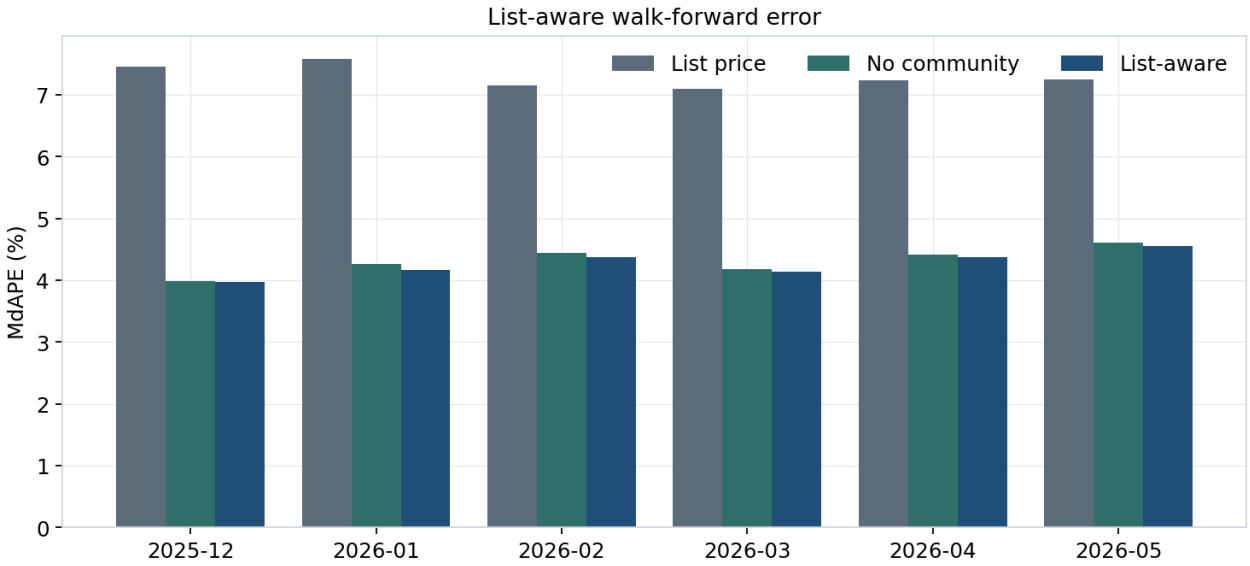


Figure 3: Walk-forward MdAPE by sale month. The list price is the baseline. The middle bar is the same model without community as a category.

C.2 Off-market

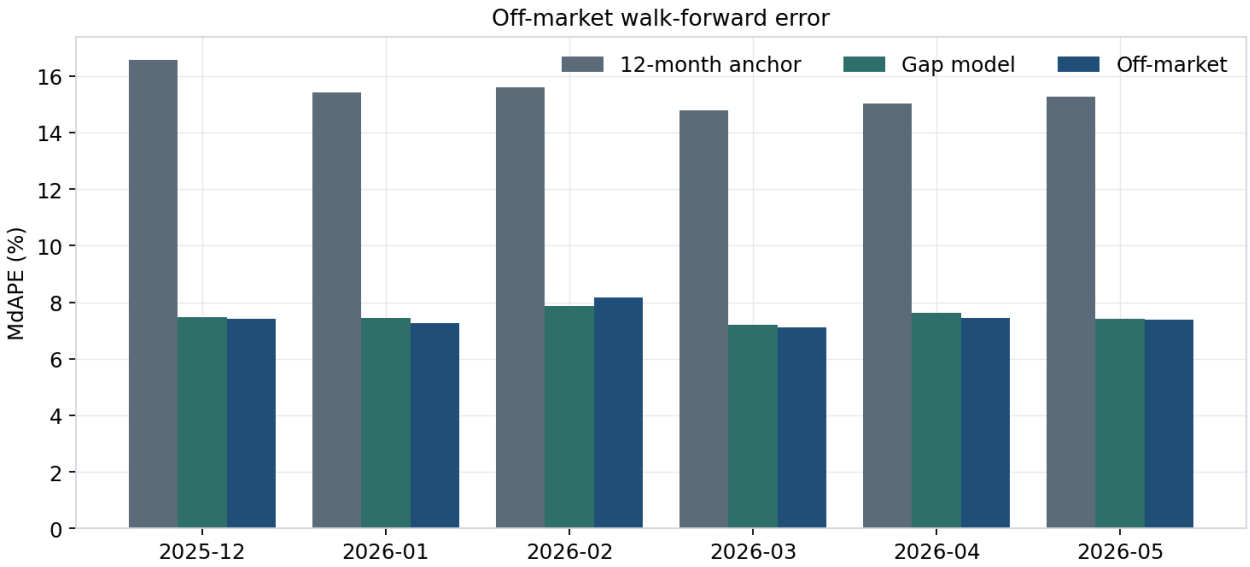


Figure 4: Walk-forward MdAPE by sale month. The 12-month community anchor is the baseline. The gap model prices the home as that anchor times a tree. February is the month the gap model is lower.

Appendix D Specifications we measured and did not keep

Choices were frozen on July–November 2025, or confirmed on the walk-forward. A win on one frozen six-month cut was not enough.

D.1 List-aware

Candidate	What happened
Remarks, text components	Developmental MdAPE 4.85 against 4.83. Left out.
A level shift after the tree	Best one-origin cell, 4.40. Walk-forward signed error −1.75. Not adopted.
Rotated coordinates, clusters, 10 nearest sales	Developmental MdAPE 4.52–4.53. No improvement.

D.2 Off-market

Candidate	What happened
Price = community anchor × a tree	Walk-forward MdAPE 7.50, against 7.45. Kept as the challenger.
Separate house and condo shifts	8.18 against 8.17 for one shift, on the inner screen.
Quantile loss at the median	Walk-forward MdAPE 7.53, behind squared error.
A land term and a drifting city forecast	The inner screen turned both off.
All-history community $/sqft	MdAPE 16.42, signed +11.97%. Stale, and high.